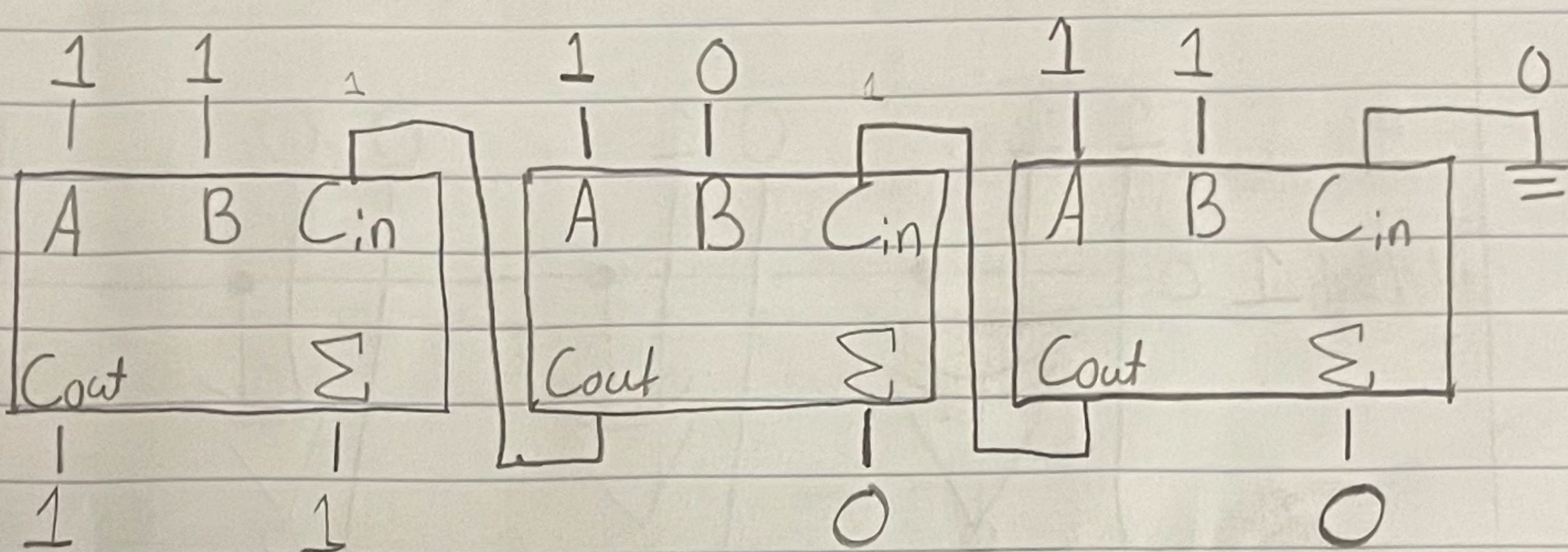


Homework 6

Ch 6: 4, 5, 7, 8, 13, 14, 19

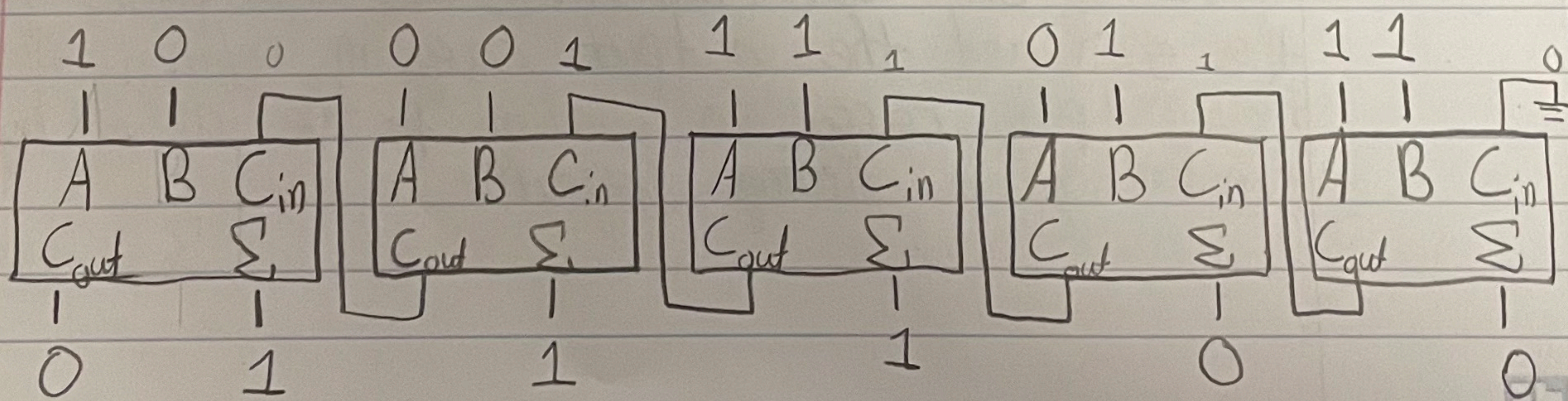
- 4 For the parallel adder in Figure 6-69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers



$$\begin{array}{r} 11 \\ 111 \\ + 101 \\ \hline 1100 \end{array}$$

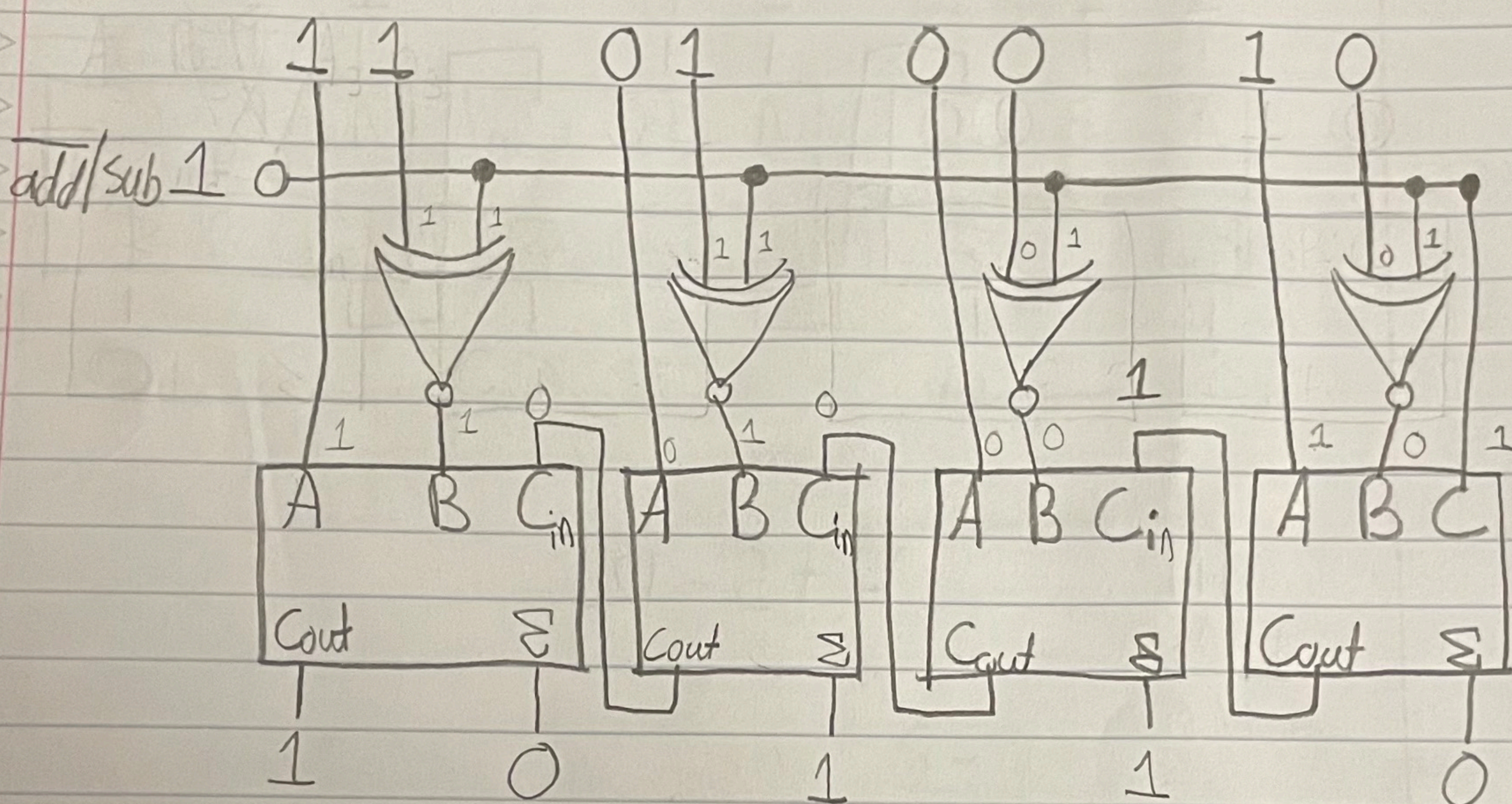
$$7 + 5 = 12$$

- 5 repeat Problem 4 for the circuit and input conditions in figure 6-70

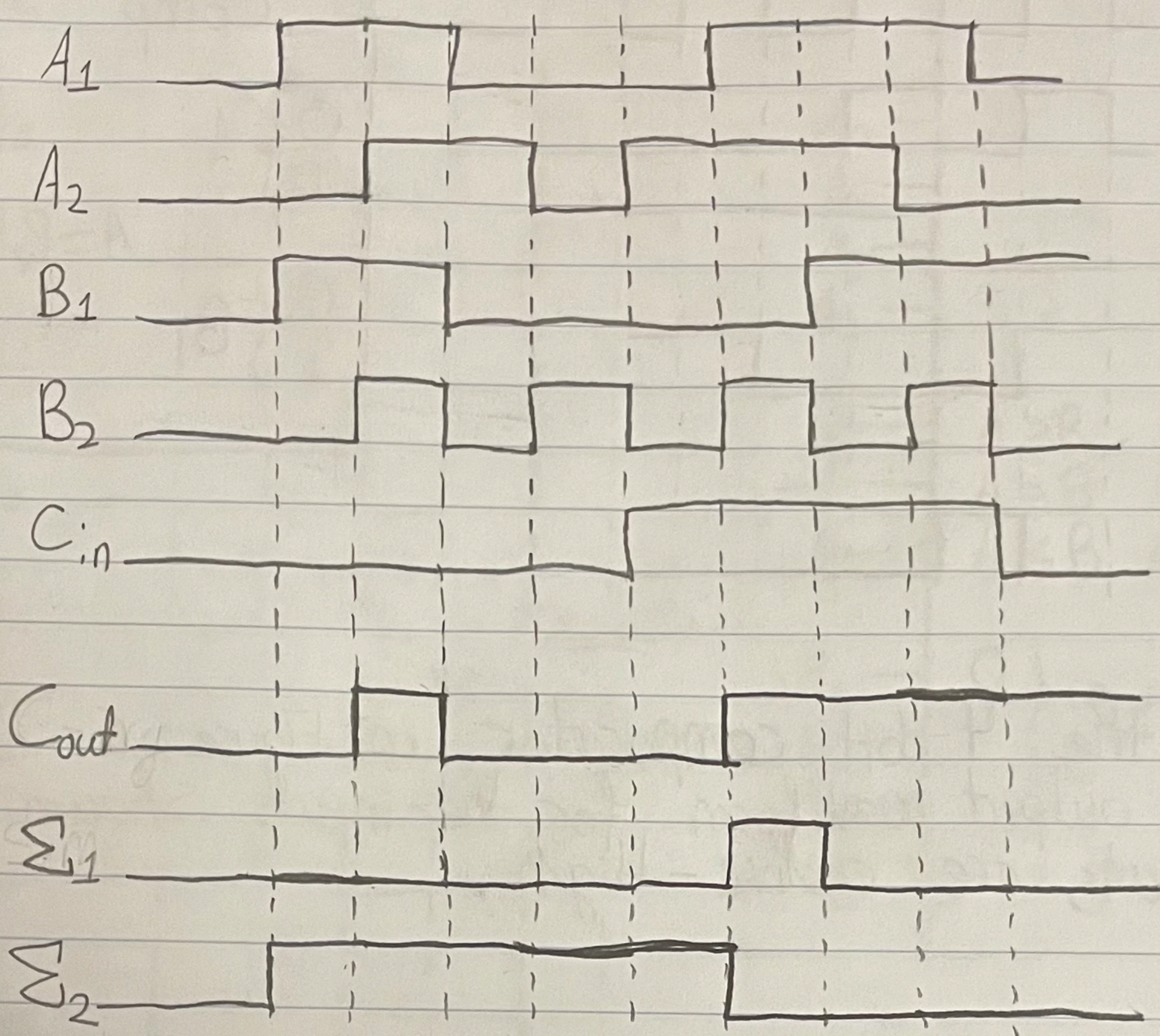


$$\begin{array}{r}
 111 \\
 10101 \\
 +00111 \\
 \hline
 11100
 \end{array}
 \quad 21 + 7 = 28$$

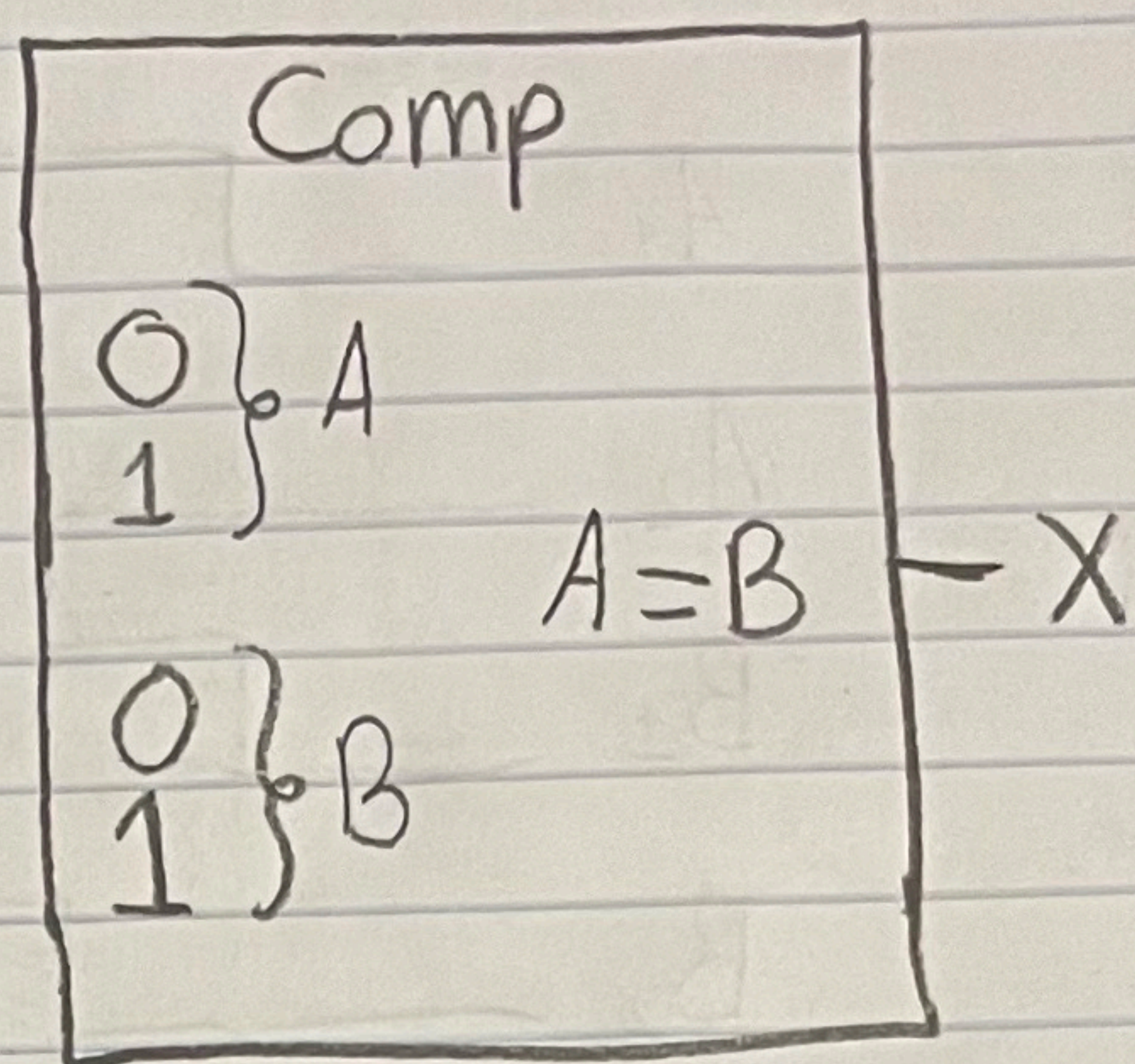
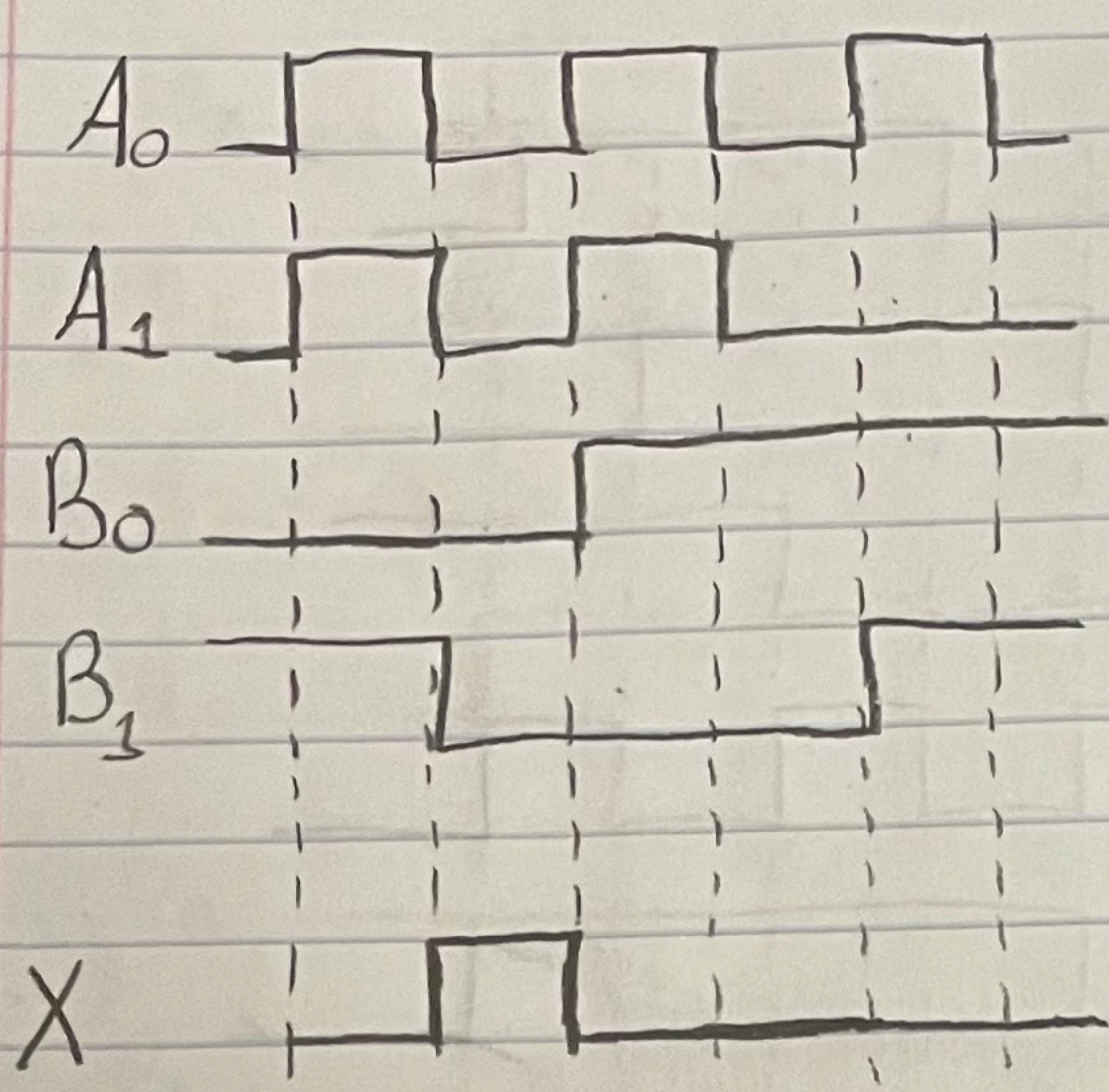
7 For the circuit in Figure 6-71, assume the inputs are Add/Subt. = 1, $A = 1001$ and $B = 1100$. What is the output?



8 The input waveforms in figure 6-72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation into the output carry in relation to the inputs by constructing a timing diagram.

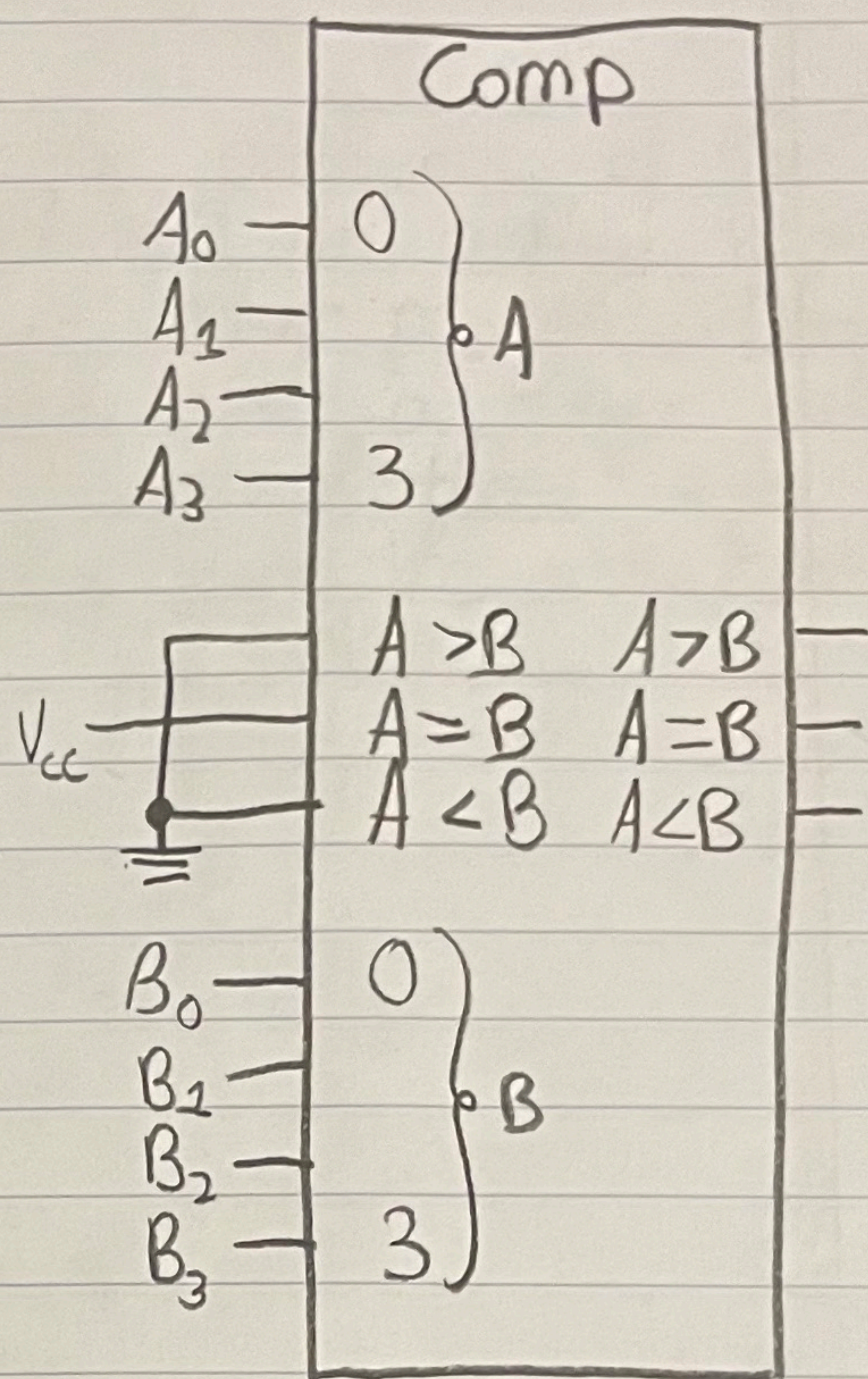
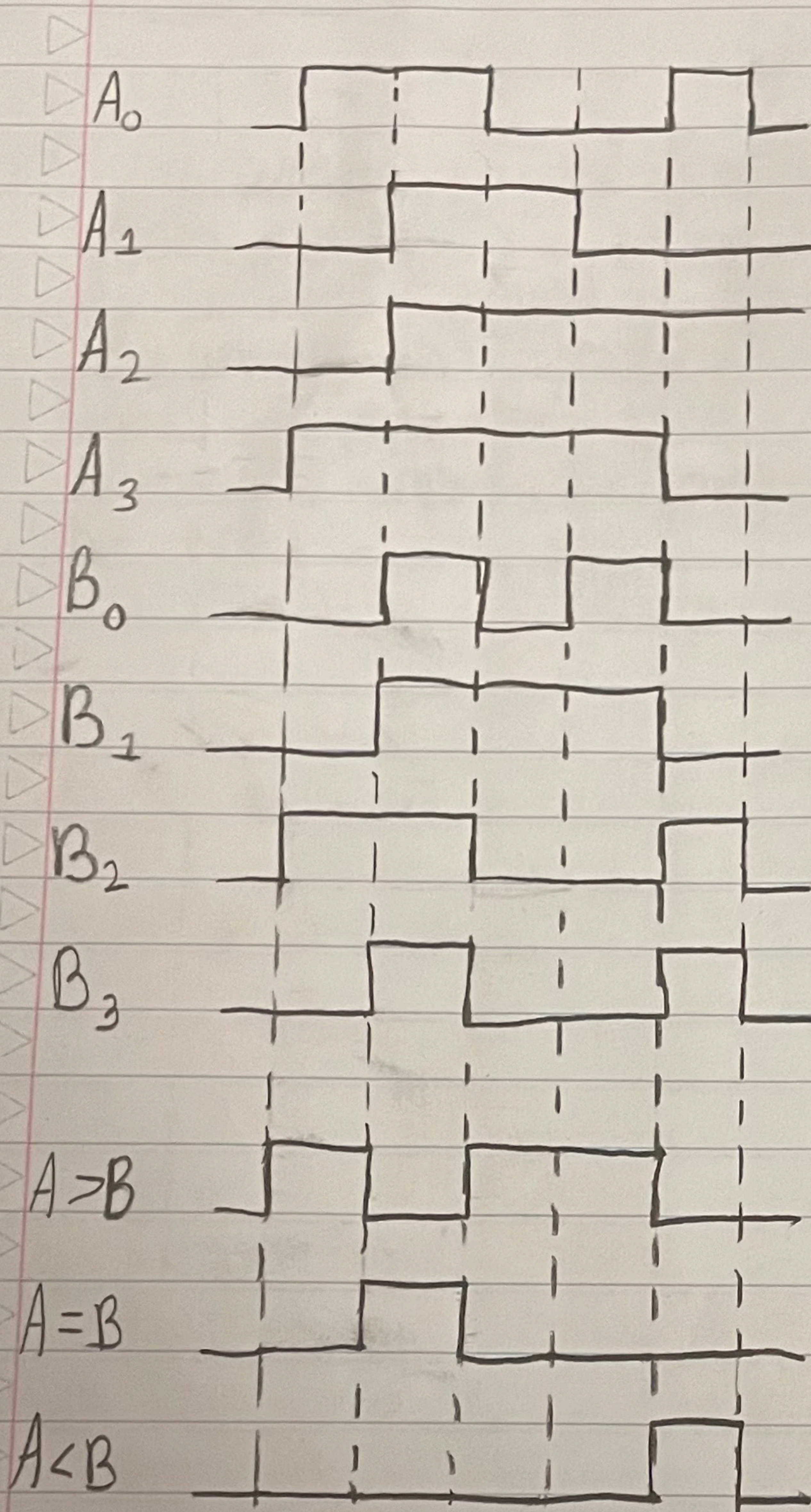


13 The waveforms in figure 6-73 are applied to the comparator as shown. Determine the output



14 For the 4-bit comparator in Figure 6-74, plot each output waveform for the inputs shown. The outputs are active-High

14



19 you wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active High output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. for any other code, the output must be LOW

A_0	A_1	A_2	A_3	X
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0